From burst to vessel graph: stabilisation, optical density and Hessian vesselness

11 October 2026

This note sets out the first stages of vessel annotation as equations: an unstabilised burst is aligned by phase correlation and averaged (section 1); the average is converted to optical density (section 2); multi-scale Gaussian derivatives give the Hessian at every pixel and scale (sections 3–4), from which Frangi's vesselness and a width estimate follow (sections 5–6); and vesselness is turned into a graph (section 7). Section 8 shows where this Hessian chain fails, which is why the later stages use oriented cells.

Notation

$I_k(\mathbf{x})$, $k = 1, \dots, N$	the frames of the burst; mono12, so values are DN in $[0, 4095]$
$\mathbf{x} = (x, y)$	pixel position; the image is $W \times H$ pixels
$M_k(\mathbf{x}) \in \{0, 1\}$	validity mask of frame k
$\mathcal{F}$, $\mathcal{F}^{-1}$	discrete Fourier transform and its inverse
G_σ	2-D Gaussian of standard deviation σ (px); $*$ is convolution
$\bar{I}(\mathbf{x})$	the stabilised average
$\mathrm{OD}(\mathbf{x})$	optical density; $\sigma_n(\mathbf{x})$ its noise

1 Burst to stabilised average

Valid pixels. A pixel is invalid where it is saturated:

$$M_k(\mathbf{x}) = \begin{cases} 0 & I_k(\mathbf{x}) = 4095 \\ 1 & \text{otherwise.} \end{cases} \tag{1}$$

Invalid pixels are filled with a local mean before any Fourier transform, so they do not ring, and are kept invalid in the average.

The Fourier shift theorem. With the discrete transform

$$\mathcal{F}\{I\}(u, v) = \sum_{x,y} I(x, y)\, e^{-2\pi i (ux/W + vy/H)}, \tag{2}$$

a translation by $\Delta = (\Delta_x, \Delta_y)$ multiplies the transform by a phase ramp:

$$I(\mathbf{x} - \Delta) \;\longleftrightarrow\; \mathcal{F}\{I\}(u, v)\, e^{-2\pi i\, \phi(\Delta)}, \qquad \phi(\Delta) = \frac{u\Delta_x}{W} + \frac{v\Delta_y}{H}. \tag{3}$$

Shift estimate by phase correlation. Window each frame with a Hann window $w(\mathbf{x})$, so the image edges do not dominate, and transform: $F_k = \mathcal{F}\{w\, I_k\}$. If frame k is the reference moved

by Δ_k, then $F_k \approx F_{\text{ref}}\, e^{-2\pi i \phi(\Delta_k)}$, and the normalised cross-power spectrum keeps only the phase ramp:

$$R_k(u,v) = \frac{F_k(u,v)\, F^*_{\text{ref}}(u,v)}{\left|F_k(u,v)\, F^*_{\text{ref}}(u,v)\right| + \epsilon} \approx e^{-2\pi i \phi(\Delta_k)}, \qquad r_k(\mathbf{x}) = \mathcal{F}^{-1}\{R_k\}(\mathbf{x}) \approx \delta(\mathbf{x} - \Delta_k). \quad (4)$$

The shift is the position of the peak, $\Delta_k = \arg\max_{\mathbf{x}} r_k(\mathbf{x})$, with indices above $W/2$ (or $H/2$) read as negative shifts. ϵ is a small constant that keeps empty frequencies from dividing by zero.

Sub-pixel precision. Fit a parabola through the peak and its two neighbours along each axis. Along x, with r_{-1}, r_0, r_{+1} the values left of, at and right of the peak,

$$\delta_x = \frac{r_{-1} - r_{+1}}{2\,(r_{-1} - 2r_0 + r_{+1})}, \quad (5)$$

and the same along y. For higher precision, refine the peak with an upsampled DFT around it (Guizar-Sicairos, Thurman and Fienup, 2008).

Applying the shift. Undo the displacement with the shift theorem (3), on the unwindowed frame:

$$I'_k(\mathbf{x}) = \mathcal{F}^{-1}\Big\{\mathcal{F}\{I_k\}(u,v)\; e^{+2\pi i\, \phi(\Delta_k)}\Big\}(\mathbf{x}) = I_k(\mathbf{x} + \Delta_k). \quad (6)$$

The mask moves with the frame (rounded to whole pixels), and the band of $|\Delta_{k,x}|$ columns and $|\Delta_{k,y}|$ rows that wrapped around the border becomes invalid: M'_k.

Reference and rejection. Use a sharp frame (or the median frame) as the first reference, then repeat the alignment against the first average. Reject frames whose correlation peak $r_k(\Delta_k)$ is low: blinks, or motion too large for one translation.

Average.

$$\bar{I}(\mathbf{x}) = \frac{\sum_k M'_k(\mathbf{x})\, I'_k(\mathbf{x})}{\sum_k M'_k(\mathbf{x})}, \qquad \bar{I}(\mathbf{x}) = \text{NaN where } \textstyle\sum_k M'_k(\mathbf{x}) = 0. \quad (7)$$

The noise falls as $1/\sqrt{n(\mathbf{x})}$, with $n(\mathbf{x}) = \sum_k M'_k(\mathbf{x})$ the number of frames that saw the pixel.

Limit. This corrects translation only. Rotation, or tissue that deforms during the burst, needs a richer motion model; the residual shows as blur in the average.

2 Optical density

Vessels absorb light, so they are dark in $\bar{I}$. By the Beer–Lambert law the light transmitted through a path of length ℓ in a medium of attenuation μ is $I = I_0\, e^{-\mu \ell}$, so

$$\text{OD}(\mathbf{x}) = \ln \frac{B(\mathbf{x}) - d}{\bar{I}(\mathbf{x}) - d} = \mu\, \ell(\mathbf{x}), \quad (8)$$

where d is the camera's dark level (its black offset) and $B(\mathbf{x})$ is the background: a smooth surface fitted to $\bar{I}$ with the vessels masked out (the mask can come from a first pass of the vesselness below).

- Vessels become positive and the background about zero.
- Uneven illumination cancels, since B and $\bar{I}$ share it.
- Across a round vessel of radius r, the path length is the chord, so the profile is $\text{OD}(u) = 2\mu\sqrt{r^2 - u^2}$ at distance u from the centreline, before the optics' blur.

The noise map follows by propagation, $\sigma_n(\mathbf{x}) \approx \sigma_{\bar{I}}(\mathbf{x})/(\bar{I}(\mathbf{x}) - d)$, or directly from the spread of OD on background pixels around its local mean. Thresholds below are in units of σ_n.

3 Scale space

Smooth the optical density with Gaussians and take derivatives with Gaussian-derivative kernels:

$$G_\sigma(\mathbf{x}) = \frac{1}{2\pi\sigma^2}\, e^{-|\mathbf{x}|^2/2\sigma^2}, \qquad L_{xx} = \frac{\partial^2 G_\sigma}{\partial x^2} * \mathrm{OD}, \quad L_{yy} = \frac{\partial^2 G_\sigma}{\partial y^2} * \mathrm{OD}, \quad L_{xy} = \frac{\partial^2 G_\sigma}{\partial x\,\partial y} * \mathrm{OD}. \tag{9}$$

- **Invalid pixels.** First fill them by normalised convolution, $\widehat{\mathrm{OD}} = \big(G_\sigma * (M \cdot \mathrm{OD})\big)/(G_\sigma * M)$, then take derivatives, and treat pixels within about 3σ of invalid ones as unreliable. Otherwise every border and gap becomes a false edge.
- **Scales.** Steps of $\sqrt{2}$: $\sigma_j = \sigma_0\, 2^{j/2}$, $j = 0, \ldots, J-1$. With $\sigma_0 = 1$ px and $J = 8$ this is $1, 1.4, 2, 2.8, 4, 5.7, 8, 11.3$ px, which covers vessel widths of roughly 2 to 30 px (section 6).

4 The Hessian and its eigenvalues

At every pixel and scale,

$$\mathcal{H}_\sigma(\mathbf{x}) = \sigma^2 \begin{pmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{pmatrix}. \tag{10}$$

The factor σ^2 is Frangi's scale normalisation: it makes responses at different scales comparable (section 6). Its eigenvalues are

$$\lambda_\pm = \sigma^2 \left[\frac{L_{xx} + L_{yy}}{2} \pm \sqrt{\left(\frac{L_{xx} - L_{yy}}{2}\right)^2 + L_{xy}^2} \right], \tag{11}$$

relabelled λ_1, λ_2 so that $|\lambda_1| \le |\lambda_2|$. For a bright vessel on the optical density, the eigenvector of λ_1 points along the vessel and that of λ_2 across it; the vessel's direction is

$$\theta = \tfrac{1}{2}\,\mathrm{atan2}\big(2L_{xy},\; L_{xx} - L_{yy}\big). \tag{12}$$

structure (on OD)	λ_1	λ_2
background	≈ 0	≈ 0
vessel (bright line)	≈ 0	strongly negative
blob	strongly negative	strongly negative
dark line (not a vessel)	≈ 0	strongly positive

5 Frangi vesselness at one scale

Two measures (Frangi, Niessen, Vincken and Viergever, 1998):

$$R_B = \frac{\lambda_1}{\lambda_2}, \qquad S = \sqrt{\lambda_1^2 + \lambda_2^2}. \tag{13}$$

R_B is near 0 for a line and near 1 for a blob; S is near 0 on flat background. The two are combined as

$$V_\sigma(\mathbf{x}) = \begin{cases} 0 & \lambda_2 > 0, \\ \exp\!\left(-\dfrac{R_B^2}{2\beta^2}\right)\left(1 - \exp\!\left(-\dfrac{S^2}{2c^2}\right)\right) & \text{otherwise.} \end{cases} \tag{14}$$

- $\beta = 0.5$ is the usual value. The 3-D filter has a third term, for plate-like structures, with no 2-D counterpart.

- Frangi sets c to half the largest S in the image, which makes the result depend on the image's content. Here c is set in noise units instead: $c = \kappa\, S_{\text{noise}}(\sigma)$, with $S_{\text{noise}}(\sigma)$ the root-mean-square S of pure noise of level σ_n at scale σ (computed once by simulation), and $\kappa \approx 3$ to start.
- On raw intensity, where vessels are dark, the sign test flips: $V_\sigma = 0$ where $\lambda_2 < 0$. In scikit-image: `skimage.filters.frangi(image, sigmas=..., black_ridges=True)`.

6 Across scales, and the width

$$V(\mathbf{x}) = \max_j V_{\sigma_j}(\mathbf{x}), \qquad \sigma^*(\mathbf{x}) = \arg\max_{\sigma_j} V_{\sigma_j}(\mathbf{x}). \tag{15}$$

For a continuous σ^*, fit a parabola in $\log\sigma$ through the best scale and its two neighbours.

Why σ^2. Take a vessel whose cross-section is a Gaussian of height A and standard deviation s_0, $f(u) = A\, e^{-u^2/2s_0^2}$. Smoothing gives $A\, s_0 (s_0^2 + \sigma^2)^{-1/2}\, e^{-u^2/2(s_0^2+\sigma^2)}$, whose second derivative at the centre is $-A\, s_0\, (s_0^2 + \sigma^2)^{-3/2}$. With the normalisation $\sigma^{2\gamma}$ the response is

$$\rho(\sigma) = A\, s_0\, \sigma^{2\gamma} \left(s_0^2 + \sigma^2\right)^{-3/2}, \qquad \frac{d\rho}{d\sigma} = 0 \implies \sigma^{*2} = \frac{2\gamma}{3 - 2\gamma}\, s_0^2. \tag{16}$$

- Frangi's $\gamma = 1$ gives $\sigma^* = \sqrt{2}\, s_0$ and a peak height $\rho(\sigma^*) = 2A/3^{3/2} \approx 0.385\, A$ that does not depend on s_0: wide and narrow vessels of the same contrast score the same.
- Lindeberg's $\gamma = 3/4$ for ridges gives $\sigma^* = s_0$, but a peak height proportional to $s_0^{-1/2}$. Only $\gamma = 1$ makes the peak height independent of the width.

Real vessels. The chord profile of a round lumen is not Gaussian, so the relation between σ^* and the radius r must be calibrated. Matching second moments gives the rough guide $s_0^2 \approx r^2/4 + \sigma_b^2$, with σ_b the optics' blur; the numerical optimum differs from it for wide, sharp lumens:

radius r (px)	blur σ_b (px)	σ^*, numerical	σ^*, moment guide $\sqrt{2(r^2/4 + \sigma_b^2)}$
4	0	3.17	2.83
4	1.5	3.62	3.54
8	0	6.57	5.66
8	1.5	6.78	6.04
15	0	12.41	10.61
15	1.5	12.54	10.82

For $r \gg \sigma_b$, $\sigma^* \approx 0.8\, r$. In practice: tabulate $\sigma^*(r, \sigma_b)$ once on synthetic profiles, and invert it to read the radius from σ^*. Below about twice the blur every vessel has nearly the same profile shape, and the width must come from the contrast (the area under the profile) instead.

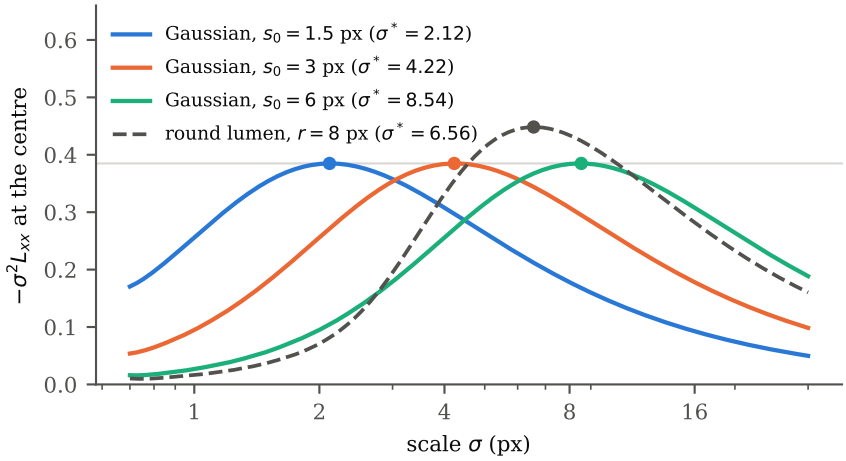


Figure 1: Scale selection with the σ^2 normalisation. Gaussian profiles of three widths peak at $\sigma^* = \sqrt{2}\, s_0$ with the same height, $2/3^{3/2}$ (grey line). A round lumen of radius 8 px (dashed, normalised to the same peak) selects $\sigma^* \approx 0.8\, r$.

7 From vesselness to a graph

This is the Hessian baseline.

1. **Centrelines.** Keep the ridge of V: pixels at least as high as their two neighbours across the vessel, $V(\mathbf{x}) \geq V(\mathbf{x} \pm \mathbf{n})$, where $\mathbf{n}$ is the unit eigenvector of λ_2 (non-maximum suppression, with sub-pixel interpolation).
2. **Hysteresis.** A ridge pixel above t_{high} starts a centreline; neighbours above t_{low} continue it.
3. **Graph.** Link ridge pixels into curves. Nodes are ends (one neighbour) and branch points (three or more). Prune spurs shorter than about one vessel width.
4. **Widths** along each centreline from σ^*, through the calibration of section 6.

8 Where the Hessian chain fails

$\mathcal{H}_\sigma$ is a symmetric 2×2 matrix: it has one pair of axes, so it can describe *one* orientation per pixel. Where two vessels meet, that is not enough. Figure 2 applies the chain above (scales 1 to 11.3 px, $\beta = 0.5$, c in noise units) to a synthetic optical-density image with three junctions.

- **Two equal vessels crossing at about 90°.** At the centre both eigenvalues are large and negative, so the pixel looks like a blob: $V = 0.20$ there, against 1.0 along the arms. The skeleton of $V > 0.5$ becomes a tangle (53 branch pixels and 6 small loops within 14 px) instead of one four-way node.
- **A fork.** The two branches stay merged in V for some distance past the split, so the skeleton's branch point lands 14 px downstream of the true fork (the trunk is 8 px wide).
- **A thin vessel crossing a wide one.** Here the topology survives (four arms leave the crossing), but inside the wide vessel V and σ^* belong to the wide vessel: the thin vessel's width and contrast are not measured where it crosses.

Oriented filters (orientation scores: one response per orientation at every pixel and scale) can hold two orientations at the same pixel, which is what junctions need. In this chain the Hessian serves as the first layer for plain vessel stretches, scale and width, and as the baseline; oriented cells take over where vessels meet.

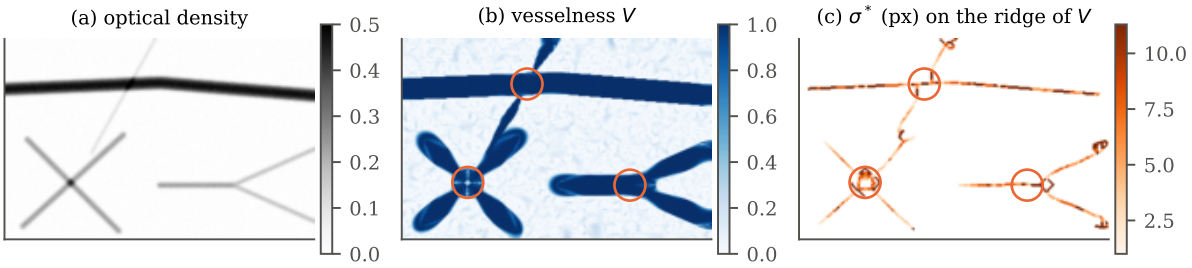


Figure 2: Frangi vesselness on a synthetic optical-density image (round lumens, blur 1.2 px, noise 0.01): a thin vessel crossing a wide one (top), two equal vessels crossing (bottom left), and a fork (bottom right). (a) The optical density. (b) Vesselness V; it falls to 0.20 at the centre of the equal crossing. (c) The selected scale σ^* on the skeleton of $V > 0.5$; wide vessels select larger scales. Circles mark the junctions.

References

- A. F. Frangi, W. J. Niessen, K. L. Vincken and M. A. Viergever. Multiscale vessel enhancement filtering. *MICCAI*, 1998.
- T. Lindeberg. Edge detection and ridge detection with automatic scale selection. *International Journal of Computer Vision* 30(2), 1998.
- M. Guizar-Sicairos, S. T. Thurman and J. R. Fienup. Efficient subpixel image registration algorithms. *Optics Letters* 33(2), 2008.